

CURRICULUM VITAE

Dr. Everardo Inzunza-González

Full Professor and Researcher (Titular C), full-time, tenured
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Profile

Everardo Inzunza-González is a full-time Professor and Researcher (Titular C) at the Faculty of Engineering, Architecture and Design of the Universidad Autónoma de Baja California (UABC) in Ensenada, Mexico, where he has worked since February 2000. He received his B.Sc. in Electronics Engineering from the Instituto Tecnológico de Culiacán, his M.Sc. in Electronics and Telecommunications from CICESE, and his Ph.D. in Electrical Sciences from UABC in 2013, with honorable mention. His recent work applies machine learning and deep learning to medical images and to biomedical signals such as EEG, ECG and heart sounds. He also keeps a line of research on chaos-based cryptography implemented on embedded systems, FPGAs and edge devices. He is a Level 2 member of Mexico's National System of Researchers (SNII, 2025–2029), has held the PRODEP Desirable Profile recognition since 2004, and leads the consolidated research group UABC-CA-238. He is part of the core faculty of the Master's and Doctoral Program in Science and Engineering (MyDCI) and has served as guest editor of special issues in MDPI and Elsevier journals.

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CITATION METRICS

Citations	h-index	i10-index
1,886	21	25

Source: Google Scholar, accessed 30 October 2026. Computed by Google Scholar over every document it attributes to the profile, including non-peer-reviewed sources; values differ from Scopus and Web of Science.

Research

Registered research lines (CVU)

- Development of intelligent systems using machine learning algorithms
- Development of systems for the productive sector

Areas with published output

- **Deep learning for medical imaging.** Brain tumor and glioma classification in MRI with CNNs and Vision Transformers; plantar footprint classification.
- **Biomedical signal processing and machine learning.** EEG for attention, motor imagery and mindfulness studies; heart sound denoising; deep vein thrombosis diagnosis.
- **Chaos-based cryptography and secure communications.** Image and ECG encryption with chaotic maps, implemented on microcontrollers, FPGAs and multiprocessor embedded systems.
- **Embedded systems and edge AI.** TinyML for gas sensing, underwater vision on ROVs, fish counting and grapevine disease detection on edge devices.
- **Robotics, path planning and logistics.** Path planning for mobile robots and drones; last-mile logistics.
- **Instrumentation and RF.** RF power amplifier linearization, de-embedding and microwave measurement calibration.

Education

2013 **Ph.D. in Electrical Sciences**
Universidad Autónoma de Baja California (UABC), Mexico
Honorable mention; academic merit award for the 2013 doctoral cohort.

2001 **M.Sc. in Electronics and Telecommunications (high-frequency track)**
Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE), Mexico

1999 **B.Sc. in Electronics Engineering**
Instituto Tecnológico de Culiacán (Tecnológico Nacional de México), Mexico

Professional experience

2000–present **Full Professor and Researcher (Titular C), full-time, tenured**
FIAD, Universidad Autónoma de Baja California
Joined as adjunct lecturer in 2000 and as full-time professor in 2001; appointed professor-researcher in November 2010.

2014–present **Chief Executive Officer**
IG Solar, Ensenada, Mexico
Photovoltaic systems company.

Research stays

2020–2021 **Sabbatical stay**
Instituto Nacional de Astrofísica, Óptica y Electrónica (INAOE) and Universidad de Guadalajara

2019–2019 **Short research stay (PRODEP support)**
INAOE
Project “Chaotic cryptography for Industry 4.0 applications”.

Academic leadership and service

Current Leader of the research group “Electronic Instrumentation Applied to Production Systems” (UABC-CA-238)
Rated as consolidated under PRODEP guidelines; evaluated as consolidated in 2019.

2015–present Core faculty member, Master's and Doctoral Program in Science and Engineering (MyDCI-UABC)

2025 Chair of the Academic Subcommittee for Electrical and Electronic Systems (SACC), MyDCI-UABC

2026–2031 Participant in the UABC–Universidade Federal de Goiás (Brazil) memorandum of understanding for an academic network in computer science and electronics

Participation in CACEI and CIEES accreditation of FIAD engineering programs and in the MyDCI evaluation before the SNP

Honors and awards

2025 **National System of Researchers (SNII), Level 2**
SECIHTI, Mexico
Term 2025–2029.

2025 **PRODEP Desirable Profile recognition**
Secretaría de Educación Pública (SEP), Mexico
Renewed for six years, 2025–2031. Previous PROMEP/PRODEP recognitions in 2004, 2007, 2010, 2013, 2016 and 2019.

2023 **PREDEPA faculty performance program, Level 8**
UABC
Terms 2021–2022, 2022–2023 and 2023–2024.

2022 **Academic Merit Award 2021, Engineering and Technology**
UABC
Presented on 25 February 2022.

2020 **National System of Researchers (SNI), Level 1**
CONACYT (now SECIHTI), Mexico

2015 **National System of Researchers (SNI), Candidate**
CONACYT (now SECIHTI), Mexico
Recorded in 2015 and 2018 in the CVU.

2015 **Third place, First National CanSat Educational Satellite Contest, project CanSat Ensenada**
University Space Engineering Consortium (UNISEC)
With Electronics Engineering students from FIAD.

2013 **Honorable mention, Ph.D. degree**
UABC

2013 **Academic merit award, 2013 doctoral cohort**
UABC

2011 **Best session paper award (Communication Systems/Technologies and Networks and their Applications I)**
10th Ibero-American Conference on Systems, Cybernetics and Informatics

2009 **Best session paper award (Communication Systems/Technologies and Networks and their Applications)**
8th Ibero-American Conference on Systems, Cybernetics and Informatics

2003 **Honorable mention for "Software educativo para el análisis y diseño de enlaces de microondas"**
Sociedad Mexicana de Instrumentación

2001 **Honorable mention for "Software educativo para calibrar el analizador de redes vectorial"**
Sociedad Mexicana de Instrumentación

Publications

42 peer-reviewed journal articles	**1** edited book (CRC Press)	**5** book chapters (Springer, CRC Press)
2 patents granted (IMPI, Mexico)	**6** software copyright registrations	**33** theses supervised or co-supervised

IEEE reference style. Authors in published order; the CV holder's name is in bold. Titles are given in their original language. Every DOI was checked against Crossref on 30 October 2026.

Journal articles (42)

42. C. Hurtado-Sánchez, R. Rosales-Cisneros, J. R. Cárdenas-Valdez, A. Calvillo-Téllez, and **E. Inzunza-González**, "Adaptive Assistive Technologies for Learning Mexican Sign Language: Design of a Mobile Application with Computer Vision and Personalized Educational Interaction," *Future Internet*, vol. 18, no. 1, Art. no. 61, 2026, doi: 10.3390/fi18010061.

41. **E. Inzunza-Gonzalez** and F. Stasolla, "Artificial Intelligence and Deep Learning in Medical and Assistive Technologies: Advances, Challenges, and Future Directions," *Technologies*, vol. 14, no. 8, Art. no. 475, 2026, doi: 10.3390/technologies14080475. EDITORIAL

40. M. A. Gómez-Guzmán, J. J. Esqueda-Elizondo, L. Jiménez-Beristain, G. M. Galindo-Aldana, O. A. Aguirre-Castro, E. R. Ramos-Acosta, C. Torres-Gonzalez, E. E. García-Guerrero, and **E. Inzunza-Gonzalez**, "Enhancing Glioma Classification in Magnetic Resonance Imaging Using Vision Transformers and Convolutional Neural Networks," *Electronics*, vol. 15, no. 2, Art. no. 434, 2026, doi: 10.3390/electronics15020434.

39. C. González-Rodríguez, E. García-Canseco, M. A. Alonso-Arévalo, **E. Inzunza-Gonzalez**, and E. E. García-Guerrero, "Enhancing heart sound signal denoising: unveiling the impact of time-frequency transformation in U-Net performance," *Signal, Image and Video Processing*, vol. 20, no. 7, Art. no. 399, 2026, doi: 10.1007/s11760-026-05421-3.

38. A. Navarro-Rodríguez, O. A. Castro-Artola, B. Rösler, E. E. García-Guerrero, O. A. Aguirre-Castro, J. Galarza-Falfan, and **E. Inzunza-Gonzalez**, "Real-time seismic signal classification using deep vision architectures for earthquake early warning systems," *Earth Science Informatics*, vol. 19, no. 4, Art. no. 44, 2026, doi: 10.1007/s12145-026-02099-6.

37. C. Lazcano-García, K. G. García-Resendiz, J. Carrillo-Tripp, **E. Inzunza-Gonzalez**, E. E. García-Guerrero, D. Cervantes-Vasquez, J. Galarza-Falfan, C. A. Lopez-Mercado, and O. A. Aguirre-Castro, "Deep Learning-Based System for Early Symptoms Recognition of Grapevine Red Blotch and Leafroll Diseases and Its Implementation on Edge Computing Devices," *AgriEngineering*, vol. 7, no. 3, Art. no. 63, 2025, doi: 10.3390/agriengineering7030063.

36. C. Hernández-Mejía, D. Torres-Muñoz, C. Maldonado-Méndez, S. Hernández-Méndez, **E. Inzunza-González**, C. Sánchez-López, and E. E. García-Guerrero, "Energy-Efficient Last-Mile Logistics Using Resistive Grid Path Planning Methodology (RGPPM)," *Energies*, vol. 18, no. 21, Art. no. 5625, 2025, doi: 10.3390/en18215625.

35. M. A. Gómez-Guzmán, L. Jiménez-Beristain, E. E. García-Guerrero, O. A. Aguirre-Castro, J. J. Esqueda-Elizondo, E. R. Ramos-Acosta, G. M. Galindo-Aldana, C. Torres-Gonzalez, and **E. Inzunza-Gonzalez**, "Enhanced Multi-Class Brain Tumor Classification in MRI Using Pre-Trained CNNs and Transformer Architectures," *Technologies*, vol. 13, no. 9, Art. no. 379, 2025, doi: 10.3390/technologies13090379.

34. A. A. Cadena Zepeda, E. E. García-Guerrero, O. A. Aguirre-Castro, G. M. Galindo-Aldana, R. Juárez-Ramírez, M. A. Gómez-Guzmán, C. Raymond, and **E. Inzunza-Gonzalez**, "Machine Learning-Based Approaches for Early Detection and Risk Stratification of Deep Vein Thrombosis: A Systematic Review," *Eng*, vol. 6, no. 9, Art. no. 243, 2025, doi: 10.3390/eng6090243.

33. G. Galindo-Aldana, L. A. Montoya-Rivera, J. J. Esqueda-Elizondo, **E. Inzunza-Gonzalez**, E. E. Garcia-Guerrero, A. Padilla-Lopez, T. G. Bautista, and C. Torres-González, "Mindfulness-Based Intervention Effects on EEG and Executive Functions: A Systematic Review," *Brain Sciences*, vol. 15, no. 3, Art. no. 324, 2025, doi: 10.3390/brainsci15030324.
Preprint version: 10.20944/preprints202503.0135.v1

32. A. Navarro-Rodríguez, O. A. Castro-Artola, E. E. García-Guerrero, O. A. Aguirre-Castro, U. J. Tamayo-Pérez, C. A. López-Mercado, and **E. Inzunza-Gonzalez**, "Recent Advances in Early Earthquake Magnitude Estimation by Using Machine Learning Algorithms: A Systematic Review," *Applied Sciences*, vol. 15, no. 7, Art. no. 3492, 2025, doi: 10.3390/app15073492.

31. L. Y. Ramírez-Rios, J. E. Olguín-Tiznado, E. R. Ramos-Acosta, **E. Inzunza-Gonzalez**, J. C. Cano-Gutiérrez, E. E. García-Guerrero, and C. Camargo-Wilson, "Toward Smarter Orthopedic Care: Classifying Plantar Footprints from RGB Images Using Vision Transformers and CNNs," *Journal of Imaging*, vol. 11, no. 11, Art. no. 414, 2025, doi: 10.3390/jimaging11110414.

30. E. R. Ramos-Acosta, E. E. García-Guerrero, O. R. López-Bonilla, U. J. Tamayo-Pérez, O. A. Aguirre-Castro, L. Y. Ramírez-Rios, and **E. Inzunza-Gonzalez**, "A novel system for the classification of zinc-plated components by benchmarking deep neural networks," *Expert Systems with Applications*, vol. 255, Art. no. 124866, 2024, doi: 10.1016/j.eswa.2024.124866.

29. M. A. Gómez-Guzmán, **E. Inzunza-González**, K. Palomino-Vizcaino, J. J. Esqueda-Elizondo, E. E. García-Guerrero, O. R. López-Bonilla, U. J. Tamayo-Perez, and L. Jiménez-Beristáin, "Analysis of immunotherapeutic control of the TH1/TH2 imbalance in a 4D melanoma model applying the invariant compact set localization method," *Alexandria Engineering Journal*, vol. 107, pp. 838–850, 2024, doi: 10.1016/j.aej.2024.09.023.

28. J. R. Cárdenas-Valdez, R. Ramírez-Villalobos, C. Ramirez-Ubieta, and **E. Inzunza-Gonzalez**, "Enhancing Security of Telemedicine Data: A Multi-Scroll Chaotic System for ECG Signal Encryption and RF Transmission," *Entropy*, vol. 26, no. 9, Art. no. 787, 2024, doi: 10.3390/e26090787.

27. J. Galarza-Falfan, E. E. García-Guerrero, O. A. Aguirre-Castro, O. R. López-Bonilla, U. J. Tamayo-Pérez, J. R. Cárdenas-Valdez, C. Hernández-Mejía, S. Borrego-Dominguez, and **E. Inzunza-Gonzalez**, "Path Planning for Autonomous Mobile Robot Using Intelligent Algorithms," *Technologies*, vol. 12, no. 6, Art. no. 82, 2024, doi: 10.3390/technologies12060082.

26. M. A. Gómez-Guzmán, L. Jiménez-Beristaín, E. E. García-Guerrero, O. R. López-Bonilla, U. J. Tamayo-Perez, J. J. Esqueda-Elizondo, K. Palomino-Vizcaino, and **E. Inzunza-González**, "Classifying Brain Tumors on Magnetic Resonance Imaging by Using Convolutional Neural Networks," *Electronics*, vol. 12, no. 4, Art. no. 955, 2023, doi: 10.3390/electronics12040955.

25. P. Góngora Lugo, E. E. García-Guerrero, **E. Inzunza-González**, E. I. Chaikina, and H. Márquez Becerra, "Desing of grating couplers for submicron optical waveguides," *Revista Mexicana de Física*, vol. 69, no. 2 Mar-Apr, 2023, doi: 10.31349/revmexfis.69.021303.

24. C. Hernández-Mejía, D. Torres-Muñoz, **E. Inzunza-González**, and C. Sánchez-López, "Exploring Robotical Implementation for Planning of Collision-free Logistical Paths Using RGPPM Algorithm," *Iranian Journal of Science and Technology, Transactions of Electrical Engineering*, vol. 47, no. 1, pp. 221–231, 2023, doi: 10.1007/s40998-022-00531-3.

23. D. A. Trujillo-Toledo, O. R. López-Bonilla, E. E. García-Guerrero, J. J. Esqueda-Elizondo, J. R. Cárdenas-Valdez, U. J. Tamayo-Pérez, O. A. Aguirre-Castro, and **E. Inzunza-González**, "Real-time medical image encryption for H-IoT applications using improved sequences from chaotic maps," *Integration*, vol. 90, pp. 131–145, 2023, doi: 10.1016/j.vlsi.2023.01.008.

22. J. R. Cárdenas-Valdez, J. A. Galaviz-Aguilar, C. Vargas-Rosales, **E. Inzunza-González**, and L. Flores-Hernández, "A Crest Factor Reduction Technique for LTE Signals with Target Relaxation in Power Amplifier Linearization," *Sensors*, vol. 22, no. 3, Art. no. 1176, 2022, doi: 10.3390/s22031176.

21. C. Hernández-Mejía, D. Torres-Muñoz, **E. Inzunza-González**, C. Sánchez-López, and E. E. García-Guerrero, "A Novel Green Logistics Technique for Planning Merchandise Deliveries: A Case Study," *Logistics*, vol. 6, no. 3, Art. no. 59, 2022, doi: 10.3390/logistics6030059.

20. J. J. Esqueda-Elizondo, R. Juárez-Ramírez, O. R. López-Bonilla, E. E. García-Guerrero, G. M. Galindo-Aldana, L. Jiménez-Beristáin, A. Serrano-Trujillo, E. Tlelo-Cuautle, and **E. Inzunza-González**, "Attention Measurement of an Autism Spectrum Disorder User Using EEG Signals: A Case Study," *Mathematical and Computational Applications*, vol. 27, no. 2, Art. no. 21, 2022, doi: 10.3390/mca27020021.

19. F. J. Ramírez-Arias, E. E. García-Guerrero, E. Tlelo-Cuautle, J. M. Colores-Vargas, E. García-Canseco, O. R. López-Bonilla, G. M. Galindo-Aldana, and **E. Inzunza-González**, "Evaluation of Machine Learning Algorithms for Classification of EEG Signals," *Technologies*, vol. 10, no. 4, Art. no. 79, 2022, doi: 10.3390/technologies10040079.

18. E. E. Contreras-Luján, E. E. García-Guerrero, O. R. López-Bonilla, E. Tlelo-Cuautle, D. López-Mancilla, and **E. Inzunza-González**, "Evaluation of Machine Learning Algorithms for Early Diagnosis of Deep Venous Thrombosis," *Mathematical and Computational Applications*, vol. 27, no. 2, Art. no. 24, 2022, doi: 10.3390/mca27020024.

17. O. A. Aguirre-Castro, E. E. García-Guerrero, O. R. López-Bonilla, E. Tlelo-Cuautle, D. López-Mancilla, J. R. Cárdenas-Valdez, J. E. Olguín-Tiznado, and **E. Inzunza-González**, "Evaluation of underwater image enhancement algorithms based on Retinex and its implementation on embedded systems," *Neurocomputing*, vol. 494, pp. 148–159, 2022, doi: 10.1016/j.neucom.2022.04.074.

16. A. Navarro-Espinoza, O. R. López-Bonilla, E. E. García-Guerrero, E. Tlelo-Cuautle, D. López-Mancilla, C. Hernández-Mejía, and **E. Inzunza-González**, "Traffic Flow Prediction for Smart Traffic Lights Using Machine Learning Algorithms," *Technologies*, vol. 10, no. 1, Art. no. 5, 2022, doi: 10.3390/technologies10010005.

15. J. A. Galaviz-Aguilar, C. Vargas-Rosales, J. R. Cárdenas-Valdez, Y. Martínez-Reyes, **E. Inzunza-González**, Y. Sandoval-Ibarra, and J. C. Núñez-Pérez, "A Weighted Linearization Method for Highly RF-PA Nonlinear Behavior Based on the Compression Region Identification," *Applied Sciences*, vol. 11, no. 7, Art. no. 2942, 2021, doi: 10.3390/app11072942.

14. L. Y. Ramírez-Rios, C. Camargo-Wilson, J. E. Olguín-Tiznado, J. A. López-Barreras, **E. Inzunza-González**, and J. L. García-Alcaraz, "Design of a Modular Plantar Orthosis System through the Application of TRIZ Methodology Tools," *Applied Sciences*, vol. 11, no. 5, Art. no. 2051, 2021, doi: 10.3390/app11052051.

13. F. Zamora-Arellano, O. R. López-Bonilla, E. E. García-Guerrero, J. E. Olguín-Tiznado, **E. Inzunza-González**, D. López-Mancilla, and E. Tlelo-Cuautle, "Development of a Portable, Reliable and Low-Cost Electrical Impedance Tomography System Using an Embedded System," *Electronics*, vol. 10, no. 1, Art. no. 15, 2021, doi: 10.3390/electronics10010015.
12. D. A. Trujillo-Toledo, O. R. López-Bonilla, E. E. García-Guerrero, E. Tlelo-Cuautle, D. López-Mancilla, O. Guillén-Fernández, and **E. Inzunza-González**, "Real-time RGB image encryption for IoT applications using enhanced sequences from chaotic maps," *Chaos, Solitons & Fractals*, vol. 153, Art. no. 111506, 2021, doi: 10.1016/j.chaos.2021.111506.
11. M. B. Fong-Mata, E. E. García-Guerrero, D. A. Mejía-Medina, O. R. López-Bonilla, L. J. Villarreal-Gómez, F. Zamora-Arellano, D. López-Mancilla, and **E. Inzunza-González**, "An Artificial Neural Network Approach and a Data Augmentation Algorithm to Systematize the Diagnosis of Deep-Vein Thrombosis by Using Wells' Criteria," *Electronics*, vol. 9, no. 11, Art. no. 1810, 2020, doi: 10.3390/electronics9111810.
10. E. E. García-Guerrero, **E. Inzunza-González**, O. R. López-Bonilla, J. R. Cárdenas-Valdez, and E. Tlelo-Cuautle, "Randomness improvement of chaotic maps for image encryption in a wireless communication scheme using PIC-microcontroller via Zigbee channels," *Chaos, Solitons & Fractals*, vol. 133, Art. no. 109646, 2020, doi: 10.1016/j.chaos.2020.109646.
9. O. A. Aguirre-Castro, **E. Inzunza-González**, E. E. García-Guerrero, E. Tlelo-Cuautle, O. R. López-Bonilla, J. E. Olguín-Tiznado, and J. R. Cárdenas-Valdez, "Design and Construction of an ROV for Underwater Exploration," *Sensors*, vol. 19, no. 24, Art. no. 5387, 2019, doi: 10.3390/s19245387.
8. A. Flores-Vergara, **E. Inzunza-González**, E. E. García-Guerrero, O. R. López-Bonilla, E. Rodríguez-Orozco, J. M. Hernández-Ontiveros, J. R. Cárdenas-Valdez, and E. Tlelo-Cuautle, "Implementing a Chaotic Cryptosystem by Performing Parallel Computing on Embedded Systems with Multiprocessors," *Entropy*, vol. 21, no. 3, Art. no. 268, 2019, doi: 10.3390/e21030268.
7. A. Flores-Vergara, E. E. García-Guerrero, **E. Inzunza-González**, O. R. López-Bonilla, E. Rodríguez-Orozco, J. R. Cárdenas-Valdez, and E. Tlelo-Cuautle, "Implementing a chaotic cryptosystem in a 64-bit embedded system by using multiple-precision arithmetic," *Nonlinear Dynamics*, vol. 96, no. 1, pp. 497–516, 2019, doi: 10.1007/s11071-019-04802-3.
6. J. M. Hernández-Ontiveros, **E. Inzunza-González**, E. E. García-Guerrero, O. R. López-Bonilla, S. O. Infante-Prieto, J. R. Cárdenas-Valdez, and E. Tlelo-Cuautle, "Development and implementation of a fish counter by using an embedded system," *Computers and Electronics in Agriculture*, vol. 145, pp. 53–62, 2018, doi: 10.1016/j.compag.2017.12.023.
5. E. Rodríguez-Orozco, E. E. García-Guerrero, **E. Inzunza-Gonzalez**, O. R. López-Bonilla, A. Flores-Vergara, J. R. Cárdenas-Valdez, and E. Tlelo-Cuautle, "FPGA-based Chaotic Cryptosystem by Using Voice Recognition as Access Key," *Electronics*, vol. 7, no. 12, Art. no. 414, 2018, doi: 10.3390/electronics7120414.
4. **E. Inzunza-González** and C. Cruz-Hernández, "Double hyperchaotic encryption for security in biometric systems," *Nonlinear Dynamics and Systems Theory*, vol. 13, no. 1, pp. 55–68, 2013. [Online]. Available: http://www.e-ndst.kiev.ua/v13n1/5(42)a.pdf
3. J. M. Hernández Ontiveros, **E. Inzunza González**, J. E. Olguín Tiznado, J. J. Soares López, P. A. Rousseau Figueroa, and C. Camargo Wilson, "802.15.4 MAC protocol applied to efficient energy consumption," *Ingeniería e Investigación*, vol. 30, no. 1, pp. 71–74, 2010, doi: 10.15446/ing.investig.v30n1.15210.
2. H. Serrano-Guerrero, C. Cruz-Hernández, R. M. López-Gutiérrez, C. Posadas-Castillo, and **E. Inzunza-González**, "Chaotic Synchronization in Star Coupled Networks of Three- Dimensional Cellular Neural Networks and Its Application in Communications," *International Journal of Nonlinear Sciences and Numerical Simulation*, vol. 11, no. 8, 2010, doi: 10.1515/ijnsns.2010.11.8.571.
1. **E. Inzunza González**, J. E. Olguín Tiznado, J. Soares López, L. Cardoza Avendaño, C. Camargo Wilson, and J. M. Hernández Ontiveros, "Developing and integrating a computer-based AM/FM radio station," *Ingeniería e Investigación*, vol. 30, no. 2, pp. 78–86, 2010, doi: 10.15446/ing.investig.v30n2.15736.

Edited book (1)

1. J. Martínez-Carranza, **E. Inzunza-González**, E. E. García-Guerrero, and E. Tlelo-Cuautle, Eds., *Machine Learning for Complex and Unmanned Systems*. CRC Press (Taylor & Francis), 2024, ISBN 978-1-032-47224-9, doi: 10.1201/9781003385615.

Book chapters (international publishers) (5)

5. X. A. Ayón-Gómez, U. J. Tamayo-Pérez, E. E. García-Guerrero, O. A. Aguirre-Castro, E. Vargas-Viveros, J. R. Cárdenas-Valdez, and **E. Inzunza-Gonzalez**, "TinyML-Oriented Workflow for VOC Classification: Breath-Acetone Case Study with Ethanol and Ammonia as Interferents," in *Complex Systems and Their Applications*, U. Uriostegui-Legorreta, E. Campos-Canton, E. Tlelo-Cuautle, and G. Huerta-Cuellar, Eds. Springer Nature Switzerland, 2026, pp. 341–358, doi: 10.1007/978-3-032-25152-7_22.
4. S. Borrego-Domínguez, L. Jiménez-Beristáin, E. E. García-Guerrero, J. J. Esqueda-Elizondo, C. Ortega-Corral, U. J. Tamayo-Pérez, D. A. Trujillo-Toledo, O. A. Aguirre-Castro, C. A. López-Mercado, and **E. Inzunza-González**, "Reinforcement Learning for Drone Path Planning: A Bibliometric Analysis," in *Complex Systems and Their Applications*, E. Tlelo-Cuautle, E. Campos-Canton, H. E. Gilardi-Velazquez, and G. Huerta-Cuellar, Eds. Springer Nature Switzerland, 2025, pp. 115–138, doi: 10.1007/978-3-031-92314-2_6.
3. O. A. Aguirre-Castro, **E. Inzunza-González**, O. R. López-Bonilla, U. J. Tamayo-Pérez, E. E. García-Guerrero, and D. A. Trujillo-Toledo, "Classifying Ornamental Fish Using Deep Learning Algorithms and Edge Computing Devices," in *Machine Learning for Complex and Unmanned Systems*, J. Martínez-Carranza, **E. Inzunza-González**, E. E. García-Guerrero, and E. Tlelo-Cuautle, Eds. CRC Press (Taylor & Francis), 2024, pp. 34–56, doi: 10.1201/9781003385615-3.
2. F. J. Ramírez-Arias, J. M. Colores-Vargas, J. O. Mérida-Rubio, E. E. García-Guerrero, and **E. Inzunza-González**, "EEG-Based Motor and Imaginary Movement Classification: ML Approach," in *Machine Learning for Complex and Unmanned Systems*, J. Martínez-Carranza, **E. Inzunza-González**, E. E. García-Guerrero, and E. Tlelo-Cuautle, Eds. CRC Press (Taylor & Francis), 2024, pp. 295–316, doi: 10.1201/9781003385615-16.
1. J. C. Castillo-García, J. E. Olguín-Tiznado, **E. Inzunza-González**, C. Camargo-Wilson, J. A. López-Barreras, and E. E. García-Guerrero, "Evaluation of the Succession Measures of the Simultaneous Perturbation Stochastic Approximation Algorithm for the Optimization of the Process Capability Index," in *Applied Decision-Making*, M. A. Sanchez, L. Aguilar, M. Castañón-Puga, and A. Rodríguez, Eds. (ser. Studies in Systems, Decision and Control). Springer International Publishing, 2019, pp. 1–26, doi: 10.1007/978-3-030-17985-4_1.

Conference papers (6)

The four WIT Press papers (2008–2011) come from the ORCID record and were verified in Crossref; they are not listed in the CVU.

6. J. Soares, C. Garcia, **E. Inzunza**, J. M. Hernandez, J. E. Olguin, C. Camargo, P. Rousseau, C. Sandoval, and A. G. Medina, "Preliminary results of landslide characteristics due to rainfall in the city of Ensenada, Baja California, Mexico using physical modelling," in *WIT Transactions on Ecology and the Environment*, 2011, pp. 623–630, doi: 10.2495/wrm110551.
5. J. Soares, R. Blanco, J. Tapia, J. R. Lerma, **E. Inzunza**, and S. Avilés, "The Baja 1000 car race and its impact on rural communities," in *WIT Transactions on Ecology and the Environment*, 2010, pp. 449–454, doi: 10.2495/st100381.
4. J. Soares, R. Blanco, C. García, **E. Inzunza**, and P. Rousseau, "Flooding tendencies for the city of Ensenada, Baja, California, Mexico, 1948–2004," in *WIT Transactions on Ecology and the Environment*, 2009, pp. 429–437, doi: 10.2495/wrm090381.
3. J. Soares, C. García, L. Mendoza, **E. Inzunza**, F. Jáuregui, and J. Obregón, "Slope instability along some sectors of the road to La Bufadora," in *WIT Transactions on Ecology and the Environment*, 2008, pp. 329–336, doi: 10.2495/st080321.
2. J. A. Reynoso-Hernandez and **E. Inzunza-Gonzalez**, "Comparison of LRL(m), TRM, TRRM and TAR, calibration techniques using the straightforward de-embedding method," in *59th ARFTG Conference Digest, Spring 2002*, 2002, pp. 93–98, doi: 10.1109/arftgs.2002.1214686.
1. J. A. Reynoso-Hernandez and **E. Inzunza-Gonzalez**, "A Straightforward De-Embedding Method for Devices Embedded in Test Fixtures," in *57th ARFTG Conference Digest*, 2001, pp. 1–5, doi: 10.1109/arftg.2001.327471.

Preprints (1)

1. G. Galindo-Aldana, L. A. Montoya-Rivera, J. J. Esqueda Elizondo, **E. Inzunza-Gonzalez**, E. E. García-Guerrero, A. Padilla, T. G. Bautista, and C. Torres-González, "Mindfulness-Based Interventions Effects on EEG and Executive Functions: A Systematic Review," *Preprints.org*, 2025, doi: 10.20944/preprints202503.0135.v1.
   Published version: 10.3390/brainsci15030324

Other articles (outreach and non-indexed venues) (13)

Classified as outreach publications in the CVU; includes a 2025 article in Revista ELECTRO listed in ORCID.

13. D. O. Roa-Rico, L. G. Trigueros-Triana, A. L. Mora-Navarro, Y. Castillo-Álvarez, L. Zamora-Rochín, G. M. Galindo-Aldana, C. Torres-González, L. A. Montoya-Rivera, **E. Inzunza-González**, L. Jiménez-Beristáin, J. J. Esqueda-Elizondo, K. V. Luengas-Bernal, and E. J. Oliveros-Martínez, "Análisis de Habilidades de Cognición Social para Evaluar el Impacto de Intervenciones Psicológicas Mediante Procesamiento Digital de Señales EEG," *Revista ELECTRO*, vol. 47, no. 1, pp. 308–313, 2025, doi: 10.67458/re2025a45308313.
12. J. R. Cárdenas-Valdez, Á. H. Corral-Domínguez, M. de J. García-Ortega, A. Calvillo-Téllez, C. Hurtado-Sánchez, and **E. Inzunza-González**, "EMG signal transmission system under RF schemes," *Pädi Boletín Científico de Ciencias Básicas e Ingenierías del ICBI*, vol. 11, pp. 277–282, 2023, doi: 10.29057/icbi.v11iespecial4.11413.
11. M. B. Fong-Mata, **E. Inzunza-González**, E. E. García-Guerrero, D. A. Mejía Medina, O. A. Morales Contreras, and A. Gómez-Roa, "Trombosis venosa profunda en extremidades inferiores: revisión de las técnicas de diagnóstico actuales y su simbiosis con el aprendizaje automático para un diagnóstico oportuno," *Revista de Ciencias Tecnológicas*, vol. 3, no. 1, pp. 23–34, 2020, doi: 10.37636/recit.v312334.
10. Y. J. Martínez-Reyes, J. R. Cárdenas-Valdez, J. A. Galaviz-Aguilar, **E. Inzunza-González**, M. de J. García-Ortega, and J. C. Núñez-Pérez, "Sistema de extracción de los coeficientes más óptimos para un modelo no lineal de un RF-PA," *Pistas Educativas*, vol. 41, no. 134, 2019. [Online]. Available: https://pistaseducativas.celaya.tecnm.mx/index.php/pistas/article/view/2062
9. J. M. Hernández-Ontiveros, **E. Inzunza-González**, E. E. García-Guerrero, and O. R. López-Bonilla, "Nuevo algoritmo para la contabilización de peces empleando momentos invariantes de Hu," *Revista Aristas: Investigación Básica y Aplicada*, 2018.
8. A. Flores-Vergara, E. E. García-Guerrero, **E. Inzunza-González**, O. R. López-Bonilla, and E. Rodríguez-Orozco, "Encriptador caótico de imágenes empleando un sistema embebido," *Revista Aristas: Investigación Básica y Aplicada*, 2017.
7. E. Rodríguez-Orozco, E. E. García-Guerrero, **E. Inzunza-González**, and O. R. López-Bonilla, "Image encryption based on improved Rössler hyperchaotic map," *Difu100ci@*, 2014. [Online]. Available: http://difu100cia.uaz.edu.mx/index.php/difuciencia/article/view/134
6. F. Zamora-Arellano, E. E. García-Guerrero, **E. Inzunza-González**, and A. Flores-Vergara, "Application of steganography and chaotic encryption to send a safety text message by email," *World Academy of Science, Engineering and Technology*, 2013.
5. C. Cruz-Hernández, **E. Inzunza-González**, R. M. López-Gutiérrez, H. Serrano-Guerrero, and E. E. García-Guerrero, "Encrypted audio communication based on synchronized unified chaotic systems," *International Journal of Electronics and Communication Engineering (WASET)*, 2010. [Online]. Available: https://publications.waset.org/13690/pdf
4. L. Cardoza-Avendaño, E. E. García-Guerrero, **E. Inzunza-González**, R. M. López-Gutiérrez, J. E. Olguín-Tiznado, and J. M. Hernández-Ontiveros, "Diseño de amplificadores de mediana potencia para aplicaciones WiMAX," *Revista Ibérica de Sistemas, Cibernética e Informática*, 2009. [Online]. Available: https://www.iiisci.org/journal/pdv/risci/pdfs/GC906OD.pdf
3. L. Cardoza-Avendaño, R. M. López-Gutiérrez, **E. Inzunza-González**, C. Cruz-Hernández, E. E. García-Guerrero, V. V. Spirin, and H. Serrano-Guerrero, "Encrypter information software using chaotic generators," *International Journal of Computer and Information Engineering (WASET)*, 2009. [Online]. Available: https://publications.waset.org/15852/pdf
2. R. M. López-Gutiérrez, E. Rodríguez-Orozco, C. Cruz-Hernández, **E. Inzunza-González**, C. Posadas-Castillo, E. E. García-Guerrero, and L. Cardoza-Avendaño, "Secret communications using synchronized sixth-order Chua's circuits," *International Journal of Electronics and Communication Engineering (WASET)*, 2009. [Online]. Available: https://publications.waset.org/15198/pdf

1. **E. Inzunza-González**, C. Cruz-Hernández, R. M. López-Gutiérrez, E. E. García-Guerrero, L. Cardoza-Avendaño, and H. Serrano-Guerrero, "Software to encrypt messages using public-key cryptography," *International Journal of Information and Communication Engineering (WASET)*, 2009. [Online]. Available: https://publications.waset.org/11798/pdf

Chapters in UABC edited volumes (15)

15. E. R. Ramos-Acosta, E. E. García-Guerrero, O. R. López-Bonilla, U. J. Tamayo-Pérez, O. A. Aguirre-Castro, and **E. Inzunza-González**, "Implementación de un sistema inteligente autónomo utilizando modelos de aprendizaje profundo," ch. 25 in an edited volume. Universidad Autónoma de Baja California, 2022, ISBN 979-8-9855972-2-6.
14. J. Galarza-Falfán, E. E. García-Guerrero, O. R. López-Bonilla, U. J. Tamayo-Pérez, O. A. Aguirre-Castro, and **E. Inzunza-González**, "Planificación de trayectorias para robot móvil autónomo en entornos industriales usando algoritmos inteligentes," ch. 26 in an edited volume. Universidad Autónoma de Baja California, 2022, ISBN 979-8-9855972-2-6.
13. E. E. García-Guerrero, L. Jiménez-Beristáin, **E. Inzunza-González**, O. R. López-Bonilla, J. J. Esqueda-Elizondo, and M. A. Gómez-Guzmán, "Evaluación de la dinámica global en modelo de cáncer cerebral usando técnicas de aprendizaje automático y método iterativo de localización," ch. 16 in an edited volume. Universidad Autónoma de Baja California, 2021, ISBN 979-8-9855972-0-2.
12. O. R. López-Bonilla, E. E. Contreras-Luján, **E. Inzunza-González**, and E. E. García-Guerrero, "Desarrollo de un nuevo instrumento electrónico portable para el diagnóstico temprano de trombosis venosa profunda empleando redes neuronales artificiales," ch. 19 in an edited volume. Universidad Autónoma de Baja California, 2020, ISBN 978-0-9998657-4-3.
11. O. R. López-Bonilla, E. E. García-Guerrero, **E. Inzunza-González**, and E. R. Ramos-Acosta, "Diseño de un sistema de monitoreo de procesos de manufactura mediante una red inalámbrica 6LoWPAN y RFID," ch. 13 in an edited volume. Universidad Autónoma de Baja California, 2020, ISBN 978-0-9998657-4-3.
10. **E. Inzunza-González**, E. E. García-Guerrero, A. Navarro-Espinoza, and O. R. López-Bonilla, "Diseño y construcción de un controlador de semáforo inteligente con comunicación inalámbrica," ch. 12 in an edited volume. Universidad Autónoma de Baja California, 2020, ISBN 978-0-9998657-4-3.
9. A. Vargas-Villavicencio, **E. Inzunza-González**, E. E. García-Guerrero, and O. R. López-Bonilla, "Evaluación de protocolos eficientes para aplicaciones en Internet de las cosas," ch. 15 in an edited volume. Universidad Autónoma de Baja California, 2019, ISBN 978-0-9998657-5-0.
8. D. A. Trujillo-Toledo, **E. Inzunza-González**, O. R. López-Bonilla, and E. E. García-Guerrero, "Encriptado caótico de imágenes digitales empleando cómputo paralelo en sistemas embebidos de alto rendimiento," ch. 8 in an edited volume. Universidad Autónoma de Baja California, 2018, ISBN 978-0-9998657-3-6.
7. J. J. Esqueda-Elizondo, **E. Inzunza-González**, J. R. Juárez-Ramírez, O. R. López-Bonilla, and E. E. García-Guerrero, "Método de medición de señales electroencefalográficas para detectar lapsos de atención en usuarios con trastorno del espectro autista," ch. 21 in an edited volume. Universidad Autónoma de Baja California, 2018, ISBN 978-0-9998657-3-6.
6. O. A. Aguirre-Castro, **E. Inzunza-González**, and E. E. García-Guerrero, "Visión artificial en vehículos acuáticos no tripulados," ch. 13 in an edited volume. Universidad Autónoma de Baja California, 2018, ISBN 978-0-9998657-3-6.
5. O. R. López-Bonilla, E. E. García-Guerrero, J. M. Colores-Vargas, **E. Inzunza-González**, and F. J. Ramírez-Arias, "Control de mecanismos por medio de señales cerebrales," ch. 13 in an edited volume. Universidad Autónoma de Baja California, 2017, ISBN 978-0-9998657-0-5.
4. **E. Inzunza-González**, E. E. García-Guerrero, and O. R. López-Bonilla, "Creación de un comité de evaluación de cesión de derechos de propiedad intelectual para fomentar el emprendimiento," ch. 15 in an edited volume. Universidad Autónoma de Baja California, 2017, ISBN 978-607-607-373-5.
3. E. E. García-Guerrero, E. Montoya-Rafael, **E. Inzunza-González**, and O. R. López-Bonilla, "Desarrollo de un dispositivo electrónico para pHmetría esofágica ambulatoria," ch. 14 in an edited volume. Universidad Autónoma de Baja California, 2017, ISBN 978-0-9998657-0-5.
2. **E. Inzunza-González**, O. A. Aguirre-Castro, and O. R. López-Bonilla, "Diseño y construcción de un prototipo electrónico submarino para aplicaciones oceanográficas," ch. 29 in an edited volume. Universidad Autónoma de Baja California, 2016, ISBN 978-0-692-82901-1.

1. **E. Inzunza-González** and M. B. Fong-Mata, "Dispositivo de ultrasonido Doppler para el diagnóstico computarizado de trombosis venosa profunda en extremidades," ch. 9 in an edited volume. Universidad Autónoma de Baja California, 2016, ISBN 978-0-692-82901-1.

Data and code in public repositories

- dtv-data — Supplementary material for "Evaluation of Machine Learning Algorithms for Early Diagnosis of Deep Venous Thrombosis" (2022).
- dataset_jj — Feature dataset for attention classification.
- eeg_features_v1 — Dataset for classifying motor imagery EEG signals with machine learning algorithms.

Intellectual property

Patents

Titles in the original Spanish.

- **Sistema y método contador automático de peces vivos usando procesamiento digital de imágenes en tiempo real**. Patent MX 385977, application MX/a/2018/002664 filed 2 March 2018. Granted (IMPI, Mexico). Holders/inventors: E. Inzunza-González, O. R. López-Bonilla, J. M. Hernández-Ontiveros, E. E. García-Guerrero.
- **Dispositivo periférico y método encriptador/desencriptador hipercaótico de imágenes digitales**. Patent MX 350648, application MX/a/2012/014313 filed 21 November 2012. Granted (IMPI, Mexico). Holders/inventors: O. R. López-Bonilla, E. E. García-Guerrero, E. Inzunza-González.

Software copyright registrations

Copyright registrations with Mexico's National Copyright Institute (INDAUTOR); the year is the application year. Titles are given in the original Spanish.

- Programa para obtención de features para medición de atención. Registration 03-2022-082210535400-01 (2022).
- Entrenamiento de sistema de modelo inteligente para predicción de la trombosis venosa profunda mediante la escala de Wells. Registration 03-2022-082210523200-01 (2022).
- Criptografía caótica en sistemas embebidos de alto desempeño. Registration 03-2019-091111224200-01 (2019).
- Diseño y construcción de un prototipo electrónico submarino para aplicaciones oceanográficas. Registration 03-2019-082611242900-01 (2019).
- Software para el encriptado de imágenes digitales, versión 1.0. Registration 03-2017-121809410300-01 (2017).
- Software para el cálculo de enlaces de microondas, versión 1.0. Registration 03-2017-121809452000-01 (2017).

Research projects

Project titles translated from the Spanish institutional records.

2026–2027 **Measuring the impact of play-based interventions on students with stress and anxiety using EEG signals and artificial intelligence**
UABC

2025–2026 **Intelligent system to support early detection of mental disorders in magnetic resonance imaging, computed tomography and EEG signals with advanced artificial intelligence techniques**
UABC, with Instituto Tecnológico de Tijuana, Instituto Tecnológico de Misantla and Universidad Tecnológica de Tijuana

2024–2025 **Randomized controlled trials to measure the neurophysiological effects (EEG, artificial intelligence and neuropsychological techniques) of a mindfulness-based intervention in young adults with anxiety and depression symptoms**
UABC (inter-faculty project)

Conference presentations

36 oral presentations, posters, full papers and keynotes recorded in the CVU. Titles in their original language.

2024 Aplicaciones de la inteligencia artificial en la clasificación de tumores cerebrales. *Congreso Multidisciplinario Internacional 2024, Inteligencia Artificial «Retos y expectativas»*, Mexico. Keynote.

2024 Neurophysiological effects study of mindfulness interventions in university students with anxiety and depression symptoms using artificial intelligence techniques: a systematic review. *EDIESCA 2024*, Mexico. Oral presentation.

2023 Path planning for industrial autonomous mobile robot using reinforcement learning algorithms. *4th Meeting for the Dissemination and Research in the Study of Complex Systems and their Applications (EDIESCA 2023)*, Mexico. Oral presentation.

2023 Complex nanostructure based on metallic nanoclusters and functionalized carbon nanotubes: an application in glucose measurement. *EDIESCA 2023*, Mexico. Oral presentation.

2023 Brain tumor classification using convolutional neural networks on MRI images. *3rd International Virtual Conference on Neuroscience, Brain Disorders and Therapeutics*, United Kingdom. Keynote.

2022 Clasificación de tumores cerebrales en imágenes de resonancia magnética usando redes neuronales convolucionales. *7.º Foro Nacional de Bioingeniería*, Mexico. Oral presentation.

2022 Real-time medical image encryption for H-IoT applications using enhanced chaotic maps. *3rd Meeting for the Dissemination and Research in the Study of Complex Systems and their Applications (EDIESCA 2022)*, Mexico. Oral presentation.

2022 Attention measurement of engineering students doing academic activities using machine learning algorithms on EEG signals. *EDIESCA 2022*, Mexico. Oral presentation.

2021 Attention measurement of an ASD user using EEG signals: a case study. *9th International Workshop on Numerical and Evolutionary Optimization (NEO 2021)*, Mexico. Oral presentation.

2021 Assembly components detection system based on convolutional neural networks: a case study. *NEO 2021*, Mexico. Oral presentation.

2021 Traffic flow prediction for smart traffic lights. *NEO 2021*, Mexico. Oral presentation.

2021 Evaluation of machine-learning algorithms for early diagnosis of deep venous thrombosis. *NEO 2021*, Mexico. Oral presentation.

2019 Diseño y simulación de un oscilador caótico en MATLAB y Verilog. *3.er Coloquio en Electrónica Analógica y Digital (COLEAD 2019)*, Mexico. Oral presentation.

2019 Trombosis venosa profunda en extremidades inferiores: revisión de las técnicas de diagnóstico actuales y su simbiosis con el aprendizaje automático para diagnóstico oportuno. *2.º Congreso Internacional de Ciencias de la Ingeniería y Tecnología*, Mexico. Oral presentation.

2019 Hardware implementation of the phase distortion to amplitude conversion algorithm applied for a 1.84-GHz PA. *7th International Workshop on Numerical and Evolutionary Optimization (NEO 2019)*, Mexico. Oral presentation.

2018 Nuevo algoritmo para la contabilización de peces empleando momentos invariantes de Hu. *IV Congreso Internacional de Investigación Tijuana – 1st Conference on Electronic Engineering (COEE)*, Mexico. Oral presentation.

2017 Sistema optoelectrónico para la contabilización de peces vivos. *39 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2017)*, Mexico. Full paper.

2016 Sistema domótico controlado por correo electrónico. *38 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2016)*, Mexico. Full paper.

2016 Encriptador y desencriptador caótico de imágenes digitales usando un sistema embebido con comunicación Wi-Fi. *ELECTRO 2016*, Mexico. Oral presentation.

2015 Implementación de un sistema criptográfico caótico en FPGA con aplicación en imágenes digitales. *37 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2015)*, Mexico. Full paper.

2015 Sistema electrónico digital de 8 bits para generación de caos. *ELECTRO 2015*, Mexico. Full paper.

2014 Implementation of an algorithm for fish counting and identification based on Hu moments for aquaculture applications. *7.º Congreso Internacional en Ciencias Computacionales*, Mexico. Full paper.

2014 Sistema automático para conteo de peces utilizando sensores infrarrojos y comunicación inalámbrica. *36 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2014)*, Mexico. Full paper.

2014 Sistema de control digital on-off por histéresis para pH con comunicación inalámbrica, para aplicaciones acuícolas. *ELECTRO 2014*, Mexico. Full paper.

2013 Doble encriptado de imágenes digitales con esteganografía sonora y cifrado caótico en comunicaciones privadas a través de un canal público. *35 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2013)*, Mexico. Full paper.

2012 Diseño e implementación de un generador de caos basado en microcontrolador de 8 bits. *LV Congreso Nacional de Física*, Mexico. Poster.

2012 Encriptado hipercaótico en un sistema de reconocimiento de voz. *LV Congreso Nacional de Física*, Mexico. Poster.

2012 Experimental network synchronization of Chua's circuits using plastic optical fiber: application to communications. *12th Experimental Chaos and Complexity Conference*, United States. Poster.

2011 Automatización del movimiento del trolley en las grúas pórtico. *33 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2011)*, Mexico. Full paper.

2011 Diversidad espacial por igual ganancia como contramedida a desvanecimientos lentos y periodo corto en un canal Rayleigh. *10.ª Conferencia Iberoamericana en Sistemas, Cibernética e Informática*, United States. Full paper.

2010 Confirmación experimental de comunicación segura de información utilizando sincronización de circuitos de Chua de cuarto orden. *LIII Congreso Nacional de Física*, Mexico. Poster.

2010 Sistema de monitorización de signos vitales. *32 Congreso Internacional de Ingeniería Electrónica (ELECTRO 2010)*, Mexico. Full paper.

2010 Sistema de supervisión de temperatura de un invernadero mediante una comunicación inalámbrica. *9.ª Conferencia Iberoamericana en Sistemas, Cibernética e Informática*, United States. Full paper.

2010 Encrypted audio communication based on synchronized unified chaotic systems. *International Conference on Computer, Electrical, and Systems Science, and Engineering (ICCESSE 2010)*, France. Oral presentation.

2009 Software to encrypt messages using public-key cryptography. *International Conference on Computer, Electrical, and Systems Science, and Engineering*, France. Oral presentation.

2006 Avances en la realización de un analizador de redes de baja frecuencia. *Congreso Tecnología, Aprendizaje y Enseñanza de la Electrónica (TAEE 2006)*, Spain. Full paper.

Teaching

Teaching in the B.Sc. in Electronics Engineering and in the MyDCI graduate program at UABC (listed in Mexico's National Graduate System, SNP). The CVU records 103 courses between 2010 and 2026.

Graduate (Master's and Doctoral Program in Science and Engineering, MyDCI-UABC)

2010–2026

- Introduction to Deep Learning
- Neural Networks
- Modular Neural Networks
- Intelligent Models
- Artificial Intelligence
- Advanced Machine Learning Models
- Pattern Recognition
- Introduction to Electronic Vision Systems
- Digital Image Processing
- Cryptography
- Instrumentation
- Analog and Digital Instrumentation
- Virtual Electronic Instrumentation
- Digital Filtering
- Software for Electrical Engineering
- Software Development for Embedded Systems
- Hardware and Software Design of Low-Power Embedded Systems
- Time Series Modeling, Analysis and Applications
- Research Activities and Research Assistantship

Undergraduate (B.Sc. in Electronics Engineering, FIAD-UABC)

2015–2025 (terms recorded in the CVU)

- Microcontroller Systems
- Microcontrollers
- Embedded Systems

Curriculum design

2025 Curriculum design of the courses Machine Learning and Deep Learning for the new B.Sc. in Data Engineering and Artificial Intelligence (plan 2026-7, Faculty of Engineering, Mexicali campus). Approved by the UABC University Council on 9 October 2025.

2017–2019 Curriculum revision of the B.Sc. in Electronics Engineering (Mexicali, Tijuana and Ensenada campuses). Approved by the UABC University Council on 17 October 2019.

Student supervision

33 theses supervised or co-supervised: 10 doctoral, 16 master's and 7 undergraduate. The year shown is the year the degree was awarded; titles are in their original language.

Doctoral theses (10)

2025 Global dynamics evaluation in brain cancer model using deep learning techniques and iterative localization method. Advisor · UABC

2023 Control of mechanisms through electroencephalographic signals. Advisor · UABC

2022 A new method for measuring attention in users with autism spectrum disorder using machine learning algorithms and EEG signals. Advisor · UABC

2022 Encriptado caótico de imágenes digitales empleando cómputo paralelo en sistemas embebidos de alto rendimiento. Advisor · UABC

2022 Evaluation of underwater image enhancement algorithms on embedded systems. Advisor · UABC

2021 Design and construction of an electrical impedance tomography system using a single board computer. Advisor · UABC

2020 Una técnica de red neuronal artificial y un algoritmo de aumentación de datos para sistematizar el diagnóstico de trombosis venosa profunda mediante el uso de criterios de Wells. Advisor · UABC

2018 Criptografía caótica en sistemas embebidos de alto desempeño. Co-advisor · UABC

2017 Encriptado hipercaótico de imágenes digitales implementado en FPGA con bioclave de acceso. Co-advisor · UABC

2017 Sistema contador de peces en tiempo real mediante procesamiento digital de imágenes utilizando un sistema embebido. Advisor · UABC

Master's theses (16)

2026 Sensor nanoestructurado de ZnFe2O4/NTC para la detección de acetona incorporando un modelo de aprendizaje profundo y TinyML. Co-advisor · UABC

2026 Desarrollo de una aplicación móvil para la detección temprana de trombosis venosa profunda mediante algoritmos de aprendizaje automático. Co-advisor · UABC

2025 Evaluación de métodos de control convencionales y no convencionales para planificación de rutas de cuadricóptero. Advisor · UABC

2024 Planificación de trayectorias para robot móvil autónomo usando algoritmos inteligentes. Advisor · UABC

2022 Diseño y construcción de un controlador de semáforo inteligente con comunicación inalámbrica. Advisor · UABC

2022 Desarrollo de un instrumento portable para el diagnóstico temprano de trombosis venosa profunda empleando redes neuronales artificiales. Advisor · UABC

2022 Sistema de detección de componentes de zinc en una línea de ensamble manual basado en redes neuronales convolucionales. Advisor · UABC

2021 Desarrollo de un sistema criptográfico multiplataforma usando el mapa caótico de Hénon y el protocolo MQTT para transmisión segura de imágenes sobre Internet. Advisor · UABC

2018 Diseño y construcción de un prototipo electrónico submarino para aplicaciones oceanográficas. Advisor · UABC

2017 Diseño y construcción de un sistema electrónico para la contabilización de peces. Advisor · UABC

2015 Diseño y construcción de un sistema electrónico de monitorización de parámetros químicos con telemetría aplicada en acuacultura. Advisor · UABC

2015 Instrumentación electrónica para aplicaciones en acuacultura. Advisor · UABC

2013 Diseño de un prototipo encriptador-desencriptador hipercaótico de audio con comunicación WiFi. Advisor · UABC

2013 Encriptado caótico basado en microprocesador con comunicación Wi-Fi. Advisor · UABC

2008 Sistema de supervisión de temperatura de un invernadero mediante una comunicación inalámbrica. Advisor · UABC

2008 Diseño y simulación de un amplificador de alta frecuencia, de mediana potencia y alta linealidad. Advisor · UABC

Undergraduate theses (7)

2023 Sensor nanoestructurado para la detección de glucosa. Co-advisor · UABC

2012 Procesamiento digital de imágenes empleando filtros espaciales. Co-advisor · UABC

2008 Acceso remoto en tiempo real al generador eólico a través de la intranet. Co-advisor · Tecnológico de Baja California (TBC Universidad)

2006 Desarrollo e integración de una estación de radio FM basada en computadora. Advisor · UABC

2005 Sistema de medición automatizado para la caracterización de dispositivos de baja frecuencia. Advisor · UABC

2005 Diseño de filtros digitales aplicados a antenas inteligentes. Advisor · UABC

2004 Temas selectos de la ingeniería electrónica. Advisor · UABC

Thesis committees and examination boards

2024 Sistema para el control de la DMT1 y monitoreo a través de RF (in progress). Thesis committee member, M.Sc., Instituto Tecnológico de Tijuana (TecNM)

2023 Investigation of supervised learning methods for power amplifiers behavioral modeling. Thesis committee and examiner, M.Sc., Instituto Tecnológico de Tijuana (TecNM)

2023 Síntesis y aplicación de nanopartículas de plata como sensor de peróxido de hidrógeno (H2O2) en la industria alimentaria. Thesis committee and examiner, B.Sc., FIAD-UABC

2023 Algoritmo de aproximaciones estocásticas con perturbación simultánea para la optimización robusta de índices de capacidad de procesos productivos. Thesis committee and examiner, Ph.D., FIAD-UABC

2022 Diseño de rejillas de acoplamiento fibra a guía de onda. Thesis committee and examiner, Ph.D., FIAD-UABC

2022 Diseño de un sistema de linealización de comportamiento para un amplificador de potencia de LTE. Thesis committee and examiner, M.Sc., Instituto Tecnológico de Tijuana (TecNM)

2020 Diseño de un sistema de modelado ponderado a la región de compresión de un RF-PA. Thesis committee and examiner, M.Sc., Instituto Tecnológico de Tijuana (TecNM)

2017 Evaluación de la experiencia de usuario mediante sensores biométricos cerebrales. Examiner, M.Sc., Faculty of Sciences, UABC

2015 Diseño e implementación de un sistema electrónico con telemetría. Thesis committee and examiner, Graduate, FIAD-UABC

Editorial work and peer review

2026 **Guest Editor**, *Technologies (MDPI)*. Special issue "Advancements in Medical and Assistive Technologies Using Artificial Intelligence and Deep Learning Techniques — 2nd Edition"

2024–2026 **Guest Editor**, *Technologies (MDPI)*. Special issue "Advancements in Medical and Assistive Technologies Using Artificial Intelligence and Deep Learning Techniques", section Assistive Technologies

2025 **Guest Editor**, *Symmetry (MDPI)*. Special issue "Studies of Symmetry/Asymmetry in Cybersecurity"

2025 **Guest Editor**, *Integration, the VLSI Journal (Elsevier)*. Special issue "Integer and Fractional-Order Complex Systems and Their Engineering Applications"

2022 **Guest Editor**, *Integration, the VLSI Journal (Elsevier)*. Special issue associated with EDIESCA 2022

2022–2023 **Guest Editor**, *Electronics (MDPI)*. Special issue "Embedded and Integrated Circuits and Systems in Real Engineering Applications"

2024 **Section Editor**, *Chaos Theory and Applications (DergiPark)*

2023 **Invited Associate Editor**, *Revista Mexicana de Ingeniería Biomédica*. Special issue "Avances recientes de la inteligencia artificial en la ingeniería biomédica" (Recent advances of artificial intelligence in biomedical engineering)

2021–2022 **Topic Editor**, *Electronics (MDPI)*

Editor of the book *Machine Learning for Complex and Unmanned Systems* (CRC Press, 2024).

Peer review

Reviewer for the following journals, with the periods recorded in the CVU.

- IEEE Transactions on Instrumentation and Measurement (2020–2023)
- IEEE Transactions on Circuits and Systems I: Regular Papers (2019–2023)
- IEEE Transactions on Circuits and Systems II: Express Briefs (2020–2021)
- IEEE Transactions on Circuits and Systems for Video Technology (2019–2021)
- IEEE Transactions on Industrial Informatics (2023)
- IEEE Access (2020–2023)
- Integration, the VLSI Journal (2018–2024)
- Engineering Applications of Artificial Intelligence (2019–2024)
- Applied Mathematics and Computation (2018–2022)
- Chaos, Solitons & Fractals (2021)
- ISA Transactions (2021–2022)
- Information Sciences (2021)
- Journal of the Franklin Institute (2022)
- Programming and Computer Software (2021–2023)
- International Journal of Circuit Theory and Applications (2022–2023)
- IET Intelligent Transport Systems (2021–2022)
- PLOS ONE (2020–2022)
- MDPI journals: Future Internet, IJERPH, Entropy, Sensors, Applied Sciences, Computers, ISPRS IJGI and others (2020–2024)
- Revista Mexicana de Ingeniería Biomédica (2022–2024)
- Revista Aristas: Investigación Básica y Aplicada (2018)

Evaluation and review panels

- **CONACYT (now SECIHTI).** Reviewer for the Innovation Incentive Program (calls 2016, 2017 and 2018), for graduate scholarships abroad (2015 and 2016), and for an FOP16 fund project, call 2021-01 (2022).
- **SEP-PRODEP.** Reviewer of Desirable Profile applications (2015 and 2021) and of new full-time faculty projects (2020).
- **UABC.** Jury member in faculty hiring competitions (2016 and 2017), reviewer for the faculty performance program PREDEPA 2017–2018, member of the technical group for the Institutional Development Plan (2023), and evaluator of outreach projects and the Entrepreneurs Expo (2014–2018).
- **Other.** Jury member, 25th National Prototypes and Entrepreneurs Contest, DGETI-CBTIS 41 (2023).

Event organization and outreach

Event organization

2022	Organizer of the 3rd Meeting for the Dissemination and Research in the Study of Complex Systems and their Applications (EDIESCA 2022); round-table panelist.
2018	Organizer of the 4th International Research Congress Tijuana.
2017	Organizer of the 34th Engineering, Architecture and Design Days (FIAD-UABC).
2016	Organizer of the 8th International Congress on Computer Sciences (CICOMP 2016) and session chair.
2015	Organizer of the Digital Systems Seminar with IPN-CITEDI.
2014	Organizer of the 21st Engineering and Architecture Days (FIAD-UABC).
2013–2023	Organizer of the 1st, 3rd, 5th, 7th, 8th and 9th Line-Following Robot Contests (FIAD-UABC).

Science outreach

2026	Talk on early brain tumor detection with deep learning, Tecnológico Nacional de México.
2025	Demonstrations of electronics engineering and artificial intelligence applications at La Noche de las Ciencias (UABC) and at UNAM; talks on brain tumor detection in MRI, TinyML on embedded devices and acute lymphocytic leukemia detection with deep learning (UABC).
2024	Talks at the Faculty of Psychology and Education, Universidad Autónoma de Querétaro; workshop at TecNM Pánuco campus; Noche de las Estrellas (UNAM).

2023	Noche de las Estrellas (UNAM), La Noche de las Ciencias (CICESE-UABC-UNAM), and AI-on-Raspberry-Pi and robotics demonstrations at FIAD.
2022	Institutional interview “Developing intelligent systems for biomedical applications” (UABC).
2021	Talk on machine learning and signal processing for biomedical engineering, Universidad Politécnica de Chiapas.
2020	Talk on the Internet of Things, First Virtual Forum on Science, Technology and Innovation of Baja California (State Council for Science and Technological Innovation of Baja California).
2013–2019	Workshops on chaotic systems, talks at INAOE and Instituto Tecnológico de Tijuana, science fairs, Baja Seafood Expo 2014, Agrobaja 2014 and the program Cimarrones en la Ciencia y la Tecnología.

Further training and languages

2026	Generative Artificial Intelligence Engineering (course, 130 h). Iberoamerican Technology Foundation.
2016	Installation of photovoltaic systems in residential, commercial and industrial settings (course, 40 h). CONOCER, Mexico.
2003	Diploma course in Automation (232 h). UABC.

Languages: Spanish (native); English.

Contact

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Google Scholar – https://scholar.google.com/citations?user=PiphRRcAAAAJ

Web of Science – https://www.webofscience.com/wos/author/record/T-6942-2019

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